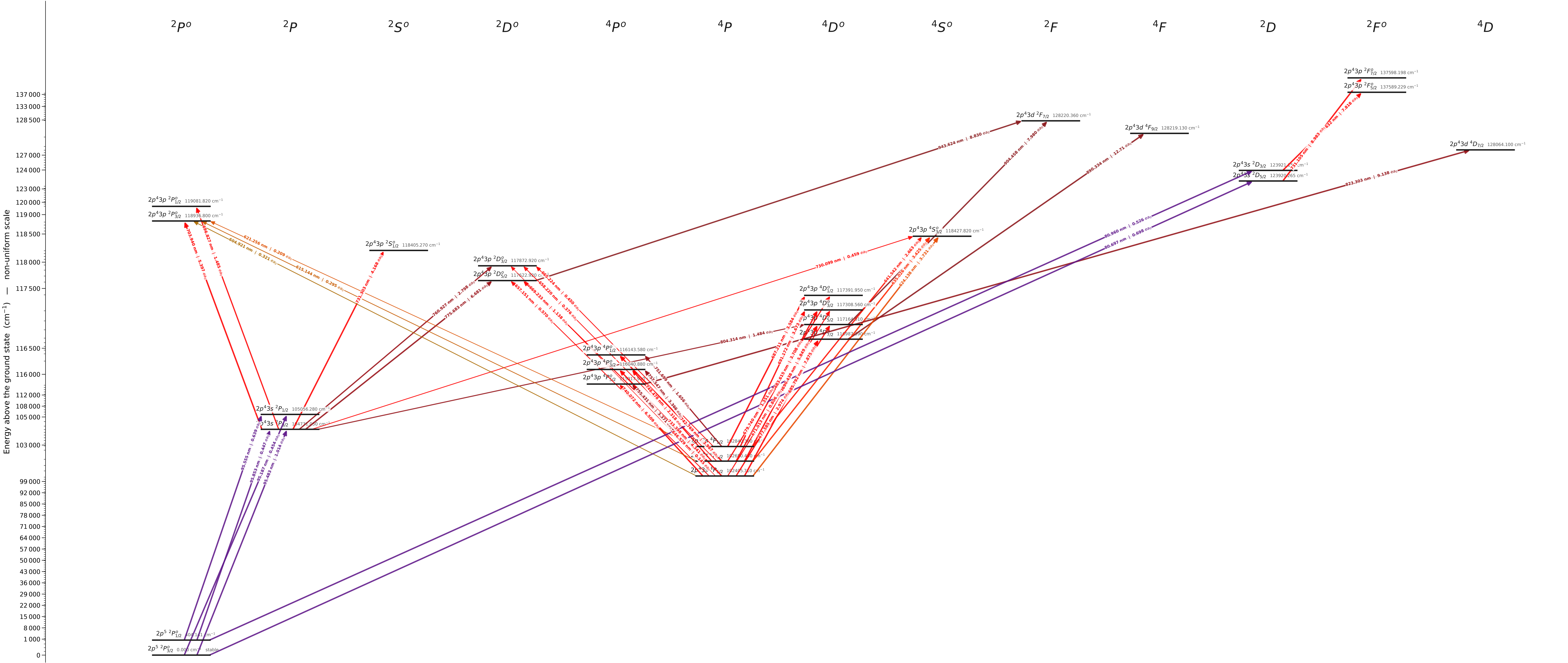
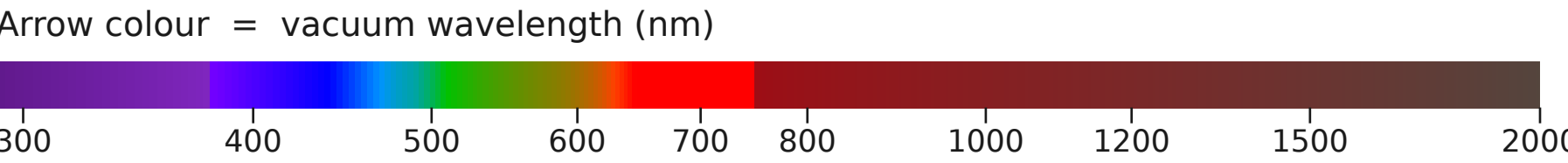


Fluorine

F (F I). Strongest classified transitions below 2 μm (NIST): 27 levels, 45 lines.

F+ ionisation limit 140524.50 cm⁻¹ = 17.42282 eV (71.162 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(U||er||J')| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	(U er J') (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2p ⁵ 2P _{3/2} - 2p ⁴ 3s 2D _{3/2}	80.6971	-	3715.0361	0.698	3.130e+08	-	-	-
2p ⁵ 2P _{1/2} - 2p ⁴ 3s 2D _{3/2}	80.9605	-	3702.9468	0.526	2.640e+08	-	-	-
2p ⁵ 2P _{3/2} - 2p ⁴ 3s 2P _{1/2}	95.1871	-	3149.5080	0.454	2.420e+08	-	-	-
2p ⁵ 2P _{3/2} - 2p ⁴ 3s 2P _{3/2}	95.4827	-	3139.7579	1.014	5.980e+08	-	-	-
2p ⁵ 2P _{1/2} - 2p ⁴ 3s 2P _{1/2}	95.5547	-	3137.3922	0.639	4.740e+08	-	-	-
2p ⁵ 2P _{1/2} - 2p ⁴ 3s 2P _{3/2}	95.8525	-	3127.6420	0.447	1.150e+08	-	-	-
2p ⁴ 3s 4P _{3/2} - 2p ⁴ 3p 4D _{3/2}	685.7920	685.6028	437.1478	7.875	4.870e+07	-	-	-
2p ⁴ 3s 4P _{1/2} - 2p ⁴ 3p 4D _{1/2}	687.2111	687.0215	436.2451	3.584	4.010e+07	-	-	-
2p ⁴ 3s 2P _{3/2} - 2p ⁴ 3p 2P _{3/2}	703.9403	703.7462	425.8777	5.397	4.230e+07	-	-	-
2p ⁴ 3s 2D _{3/2} - 2p ⁴ 3p 2F _{7/2}	731.1046	730.9032	410.0541	8.983	5.230e+07	-	-	-
2p ⁴ 3s 2P _{3/2} - 2p ⁴ 3p 2S _{1/2}	731.3031	731.1017	409.9428	4.168	4.500e+07	-	-	-
2p ⁴ 3s 2D _{3/2} - 2p ⁴ 3p 2F _{5/2}	731.6318	731.4303	409.7586	7.818	5.270e+07	-	-	-
2p ⁴ 3s 4P _{3/2} - 2p ⁴ 3p 4P _{5/2}	740.0719	739.8681	405.0856	6.509	3.530e+07	-	-	-
2p ⁴ 3s 4P _{3/2} - 2p ⁴ 3p 4P _{1/2}	742.7688	742.5642	403.6148	3.987	3.930e+07	-	-	-
2p ⁴ 3p 4P _{3/2} - 2p ⁴ 3d 4D _{1/2}	823.3034	823.0772	364.1336	9.138	3.790e+07	-	-	-
2p ⁴ 3p 4D _{3/2} - 2p ⁴ 3d 4F _{3/2}	890.3340	890.0896	336.7191	12.71	4.640e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).